Feedforward NNs continued

$X = \{ [0,0], [0,1], [1,0], [1,1] \}$ } XOR function
$\quad\quad -1 \quad +1 \quad +1 \quad -1$

$y = \{\pm 1\}$

FFNN $p(y|x) = \dfrac{e^{v(y)\cdot\phi(x;\theta)+\gamma_y}}{\sum_{y'} e^{v(y')\phi(x;\theta)+\gamma_{y'}}}$

we say this correctly models XOR if it gives higher prob to correct label

$p(y|x) > 0.5 \quad \forall$ of XOR x, y pairs

(i.e. using arg max would get a correct label)

Thm 1 if $\phi(x;\theta) = x$

then $\nexists$ rest of params $(v(y), \gamma_y)$

$= \{v(+1), v(-1), \gamma_{+1}, \gamma_{-1}\}$

that would model XOR

key lemma (general version for any $\phi(x;\theta)$)

$p(+1|x) > 0.5$

(last class, for version where

$\Leftrightarrow \quad u \cdot \phi(x;\theta) + \gamma > 0$

$\phi(x;\theta) = x$, $u \cdot x + \gamma > 0$ $\Leftrightarrow$ $p(+1|x) > 0.5$

$u := v(+1) - v(-1) \qquad \gamma = \gamma_{+1} - \gamma_{-1}$

also $P(-1|x) > 0.5 \Leftrightarrow u \cdot \phi(x;\theta) + \gamma < 0$

Thm 2 if $\phi(x;\theta)$ is a single layer FF NN with just $m=2$ neurons, and $g = \text{ReLU}$ then $\exists$ exist $v(+1), v(-1), \gamma_{+1}, \gamma_{-1}$ and θ s.t. the model models XOR.

Pf $W = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ ← w_1, ← w_2 $\qquad b = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$ ← b_1, ← b_2

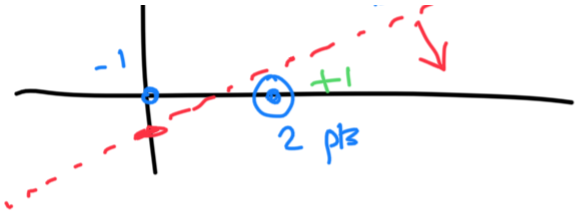
	inputs to the 2 neurons $Wx+b$	← output of neurons $h = \text{ReLU}(Wx+b)$	y
$[0,0]$	$[0,-1]$	$[0,0]$	-1
$[1,0]$	$[1,0]$	$[1,0]$	$+1$
$[0,1]$	$[1,0]$	$[1,0]$	$+1$
$[1,1]$	$[2,1]$	$[2,1]$	-1

the rest of problem is to chose v, γ

$\equiv$ same as separating plane of the 4 points $h(x^i)$



by visinspection, something w/ - bias (γ) ... separating plane



u : normal of separating plane

u : [middly pos, more neg]

By lemma need u, γ s.t

$$u \cdot [0,-1] - \gamma < 0$$

$$2,3 \rightarrow \quad u \cdot [1,0] - \gamma > 0$$

$$u \cdot [2,1] - \gamma < 0$$

$u = [1, -2]$

$\gamma = -0.5$ ✓

Choose any $v(+1), v(-1), \gamma_{+1}, \gamma_{-1}$ s.t

$$u = v(+1) - v(-1), \quad \gamma = \gamma_{+1} - \gamma_{-1}$$

Gradients / learning of FFNN

Learning $\equiv$ maximize probability of training corpus $C = \{(x_i, y_i)\}$

$\Leftrightarrow$ minimizing neg. log of prob

$$Pr[C] = \prod p_\theta(y_i | x_i)$$

by monotonicity of log, same as maximizing

$$\log Pr[C] = \sum \log p_\theta(y_i | x_i)$$

$\Leftrightarrow$ min. neg of this

ll want to min $L(\theta, v, \gamma) =$

$$-\sum \underbrace{\log p(y_i | x_i ; \theta, V, \gamma)}$$

$$:= L(x_i, y_i | \theta, V, \gamma)$$

no closed form solutions. Instead gradient methods used

Gradient method

$$\theta_j^{t+1} = \theta_j^t - \eta^t \frac{\partial L(\theta, V, \gamma)}{\partial \theta_j}$$

next classes: how to calculate these

η^t eta learning rate at time t

(also for V, γ, overall you want:

$$\frac{\partial L(\theta, V, \gamma)}{\partial \theta_j}, \quad \frac{\partial L(\theta, V, \gamma)}{\partial \gamma_y}, \quad \frac{\partial L(\theta, V, \gamma)}{\partial V_k(y)}$$

$\forall j \in |\theta|$ $\quad \forall y$ $\quad \forall y, \forall k \in \{1, \dots, m\}$

Q1 1a) $\phi(x; \theta) = g(Wx + b)$

$g(z) = a \cdot z + c$

equiv to $\phi(x; \theta) = x$

WRL

$$v'(y) = v(y) \cdot (aW) = (aW)^T v(y)$$

$= W^T(\alpha v(y))$

$\gamma_y' = v(y)\cdot \alpha b + \gamma_y + v(y)\cdot c$

$= (\alpha b + c)\cdot v(y) + \gamma_y$

1b) $v'(y) = v(y)(AW) = (AW)^T v(y)$

$= W^T A^T v(y)$

$\gamma_y' = v(y)\cdot(Ab + c) + \gamma_y$

1c) theorem from class, $\nexists\ v'(y), \gamma_y'$ when $\phi(x;\theta) = x$

but by 1a, 1b, $g(z) = \alpha\cdot z + c$ or $Az + c$ has that form □

(Q2)

2a) $\dfrac{\partial LS_y(l)}{\partial l_y}$

$= 1 - \dfrac{e^{l_y}}{\sum_{y'} e^{l_{y'}}}$

$= 1 - \text{SOFTMAX}_y(l)$

prob of label y

rate of change of label y wrt score of that label (l_y)

for huge l_y, $e^{l_y}/\sum_{y'}$ close to 1, doesn't improve

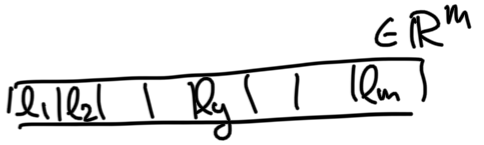
training set prob

$\prod p(y_i|x_i)$

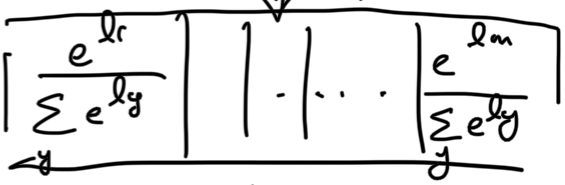
$\Downarrow$

$\sum \log p(y_i|x_i)$

scores: $l \in \mathbb{R}^m$



$\Downarrow$ SOFTMAX



$\Downarrow$ log of SOFTMAX

$LS(l)$



$$\frac{\partial L_y(l)}{\partial l_{y'}} = \frac{-e^{l_{y'}}}{\sum_{y''} e^{l_{y''}}} = -\text{SOFTMAX}_{y'}(l)$$

if SOFTMAX$_{y'}$ = 0.0001, ys prob much more

always neg: increasing another label (y')'s score can only make the ratio of $e^{l_y} / \sum_{y''}$ smaller

2b) Homework!

do $\frac{\partial l_{y'}}{\partial h_k}$ first, then $\frac{\partial l_{y'}}{\partial W_{j,k}}$